

VAIBHAV VINOD INGLE

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PROFESSIONAL SUMMARY

Civil Engineer with an MS in Civil and Construction Engineering (Research) from Oregon State University (expected Sep 2026), a Master of Technology (M.Tech) in Structural Engineering, and a Bachelor's in Civil Engineering, backed by 2 years of hands-on R&D experience in construction materials and structural systems. Experience includes mix design and performance evaluation of cement, SCM's, UHPFRC, fibers (steel, hemp), biochar, 3D-printed concrete, and soil-cement systems, with testing aligned to ASTM/EN/ISO standards and NDT techniques. Strong foundation in structural behavior, fresh/hardened-state material characterization, and translating experimental findings into practical engineering recommendations. Author of 5 peer-reviewed publications (3 first-author), reflecting technical rigor and strong scientific communication skills. Seeking a full-time Civil/Structural Engineer role to apply this multi-degree academic background and applied research experience to real-world construction and infrastructure projects.

EDUCATION

MS in Civil and Construction Engineering

Sep 2024 - Expected Sep 2026

Oregon State University, Corvallis, OR

- Supervisor: Dr. Pavan Akula | Focus: Sustainable construction materials, mix design, and additive construction (3D Printing)

M.Tech in Structural Engineering | CGPA: 7.99/10

2020 - 2022

Shri Guru Gobind Singhji Institute of Engineering and Technology (SGGSIE&T), Nanded, India

B.E. in Civil Engineering | CGPA: 8.91/10

2014 - 2018

Prof. Ram Meghe College of Engineering & Management, India

RESEARCH & PROFESSIONAL EXPERIENCE

Graduate Research Assistant

Sep 2024 - Present

Oregon State University - Civil & Construction Engineering | Corvallis, OR

- Executing PhD-level research on soil-cement-based 3D printing (extrusion additive construction) - natural soil and cement generating data on mix design, fresh-state rheology, and geotechnical performance.
- Project 1: Developing and optimizing soil-binder mixture designs for 3D-printed infrastructure, funded by ERDC;
- Project 2: Developing and optimizing soil-binder-biochar and hemp fibre mixture designs, conducting prototype design for improved acoustic and thermal performance of mass timber components using forestry waste.
- Formulating research hypotheses, designing controlled experiments, and independently managing all stages of the research outline from material selection to data analysis and scientific writing.
- Synthesizing literature across cement chemistry, rheology, admixture influence, hardened state mechanics, and additive manufacturing to position soil-cement 3D printing as a sustainable construction technology.
- Targeting 1 journal submission from MS thesis; expected graduation September 2026.

Junior Research Fellow (JRF)

Nov 2022 - Aug 2024

CSIR-Structural Engineering Research Centre (CSIR-SERC) | Chennai, India

- Designed and evaluated multi-component UHPFRC systems incorporating portland cement, silica fume, fly ash, and limestone powder as supplementary cementitious materials (SCMs) - investigating pozzolanic reactivity, particle packing, and the effect of SCM type and dosage on hydration kinetics, microstructural development, and mechanical performance.
- Optimized superplasticizer dosage and type in UHPFRC mix designs to control workability, setting behavior, open time, and buildability - developing hands-on understanding of polymer-cement interactions and admixture compatibility with blended binder systems.
- Systematically investigated admixture-free printable concrete to establish baseline cement hydration and fresh-state behavior, then advanced to superplasticizer-optimized UHPFRC - building a mechanistic, ground-up understanding of how chemical admixtures modify cement hydration and rheology.
- Pioneered Acoustic Emission (AE) characterization of 3D-printed UHPFRC during fresh-to-hardened state transitions, characterizing microcrack evolution, fracture energy, and phase development in cementitious composites; published in Journal of Building Engineering (2025).
- Conducted fresh-state testing and hardened-state testing (compressive strength, flexural performance, fracture mechanics) per ASTM and EN standards; communicated results and technical recommendations to senior scientists and project stakeholders.
- Generated 5 publications (3 first-authored) and 30 citations in around 2 years; presented at FraMCoS-XI international fracture mechanics conference.
- Supervised and coordinated junior lab staff during experimental work, ensuring data quality and protocol adherence.

Project Associate - I

Aug 2022 - Oct 2022

CSIR-Central Building Research Institute (CSIR-CBRI) | Roorkee, India

- Contributed to 3D concrete printing R&D at a leading national laboratory.

Internship during M.Tech

Jan 2022 - Jul 2022

CSIR-Structural Engineering Research Centre (CSIR-SERC) | Chennai, India

- Developed and validated printable concrete mix designs, rigorously characterized fresh and hardened-state performance to map the baseline behavior of cement systems - providing the mechanistic foundation for subsequent superplasticizer optimization work (Structural Concrete, 2023/2024).
- Conducted testing per ASTM and EN standards; communicated results and technical recommendations to senior scientists to guide next-phase mix design decisions.

Teaching Assistant - Strength of Materials

Feb 2021 - Sep 2021

SGGSJET Nanded | Nanded, India

- Designed and delivered instructional content for undergraduate civil engineering students.

PUBLICATIONS | GOOGLE SCHOLAR:

- Ingle, V.V. et al. (2023/2024). 3D Printable Concrete Without Chemical Admixtures: Fresh and Hardened Properties. Structural Concrete. <https://doi.org/10.1002/suco.202300267>
- Ingle, V.V., Prem, P.R. et al. (2025). Acoustic Emission Characterization of 3D-Printed Ultra-High-Performance Concrete Under Fresh and Hardened States. Journal of Building Engineering. <https://doi.org/10.1016/j.jobbe.2025.113491>
- Prem, P.R., Ingle, V.V. et al. (2024). Influence of Loading Rate on Flexural Performance and AE Characteristics of UHPC. Structural Engineering and Mechanics, 89(6), 617–626. <https://doi.org/10.12989/sem.2024.89.6.617>
- Ingle, V.V. et al. (2023). AE Characterization of 3D-Printed UHPC Beams Under Bending. FraMCoS-XI. <http://dx.doi.org/10.21012/FC11.092372>
- Ramamurthy, V.K., Ingle, V.V. et al. (2023). Fracture Characterization of 3D-Printed UHPFC Beams Using AE. FraMCoS-XI. <http://dx.doi.org/10.21012/FC11.092368>

AWARDS & HONORS

Travel Fellowship - ASU-Swinburne Workshop on Advances in Concrete 3D Printing

Jun 2025

Jointly awarded by Arizona State University & Swinburne University of Technology | NSF AccelNet / 3DConcrete

- Competitively selected to present research at an international workshop gathering world-leading researchers and industry professionals in 3D concrete printing (Hawthorn, Victoria, Australia, 11-13 Jun 2025); awarded USD \$2,750 travel fellowship.

Kiewit Center Fellowship

Sep 2024

School of Civil & Construction Engineering, Oregon State University

- Awarded the 2024-2025 Kiewit Center Fellowship for academic excellence and research potential; covers full Year-1 tuition, monthly stipend, and medical insurance.

GATE Post-Graduate Stipend (AICTE-MoE Scholarship)

Jan 2021

All India Council for Technical Education (AICTE) & Ministry of Education | SGGSJET Nanded

- National GATE Post Graduate Scholarship awarded to GATE (Graduate Aptitude Test in Engineering)-qualified M.Tech students at recognized institutions.

SKILLS & COMPETENCIES

Technical

• Cement Chemistry • SCM Systems (Silica Fume, Fly Ash, Limestone Powder, GGBS) • Pozzolanic Reactivity & Blended Binder Design • Chemical Admixture Science (Superplasticizers, Retarders, Accelerators) • UHPFRC / UHPC Mix Design • Fresh-State and Hardened-State Testing (Compressive, Flexural, Fracture Mechanics) • Acoustic Emission Technique (AET) • ASTM / EN / ISO Concrete & Cement Standards • 3D Concrete Printing • Soil-Cement Characterization • Microstructural Analysis (XRD, TGA and FTIR - in progress, OSU 2026) • AutoCAD • Origin Pro • Overleaf (LaTeX) • MS Office Suite

Professional

Independent Research Design & Execution • Technical Report Writing & Scientific Publication • Data Analysis & Results Communication to Technical/Non-Technical Audiences • Lab Team Coordination & Supervision • Cross-functional Collaboration • Conference Presentation

Languages

English (Fluent) • Hindi (Fluent) • Marathi (Native)

CERTIFICATIONS/LICENSURE/MEMBERSHIP

FE Exam (Fundamentals of Engineering): Intend to take upon securing employment, results available within 3 months of start date, if required.

ASCE: Society Student Membership (Member No.: 000012639707), Structural Engineering Institute Membership (SEI).